Java Programming Lab-2

1. Write a program to read and print n elements to a one dimensional array
2. Write a program to print the sum and average of the elements of an array
3. Write a program to find the maximum element in an array
4. Write a program to search for an element in an array
5. Write a program to sort an array.
6. Write a program to read and print the elements of a 2 dimensional array
7. Create a class Employee with members empid, empname, designation, salary. Also define two methods to enter the details and display the details. Create an object and invoke the methods.
8. Define a class Student with data members rollno, name, mark1, mark2, mark3, total, avg. Use appropriate methods for entering the details and displaying the details. Also define a method for calculating the total mark and average. Create an object for the class and invoke all the methods.
9. Create a class named Rectangle with data members length and breadth. Create methods for entering the details and displaying the details. Create two more methods, one for finding the area and another for finding the perimeter of the rectangle. Create an object for the class and invoke all the methods.
10. Build a class Emp which contain the following details about the employee and make an object in the main method and call the methods to set values and print values of the object.
    a. Data Members
        i. int empCode
        ii. String name
        iii. int basicPay
        iv. float DA

v. float HRA

vi. float grossPay

b. Methods

i. to accept and set values for empCode, name, and basicPay

ii. to calculate grossPay

iii. to print all the values of the employee object

If basicPay < 3500 then DA is 45% of basicPay and HRA is 20% of basicPay.

If basicPay >=3500, then DA = 50% of basicPay and HRA is Rs 1000.

Find and print the grossPay = basicPay +DA+HRA.

11. Create a class named Book with data members title, author and price. Write a no argument constructor which initializes the instance variables with some values and a parameterized constructor also to initialize variables with user input. Write a function for displaying the details. Create two objects for the class using two constructors and invoke the display function.
12. Write a Java program to overload an add function that can add integer, float or double values and display the sum.